

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q17-0866

Pana 138 kV
November 19, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0866. The Interconnection Customer proposes a 63.5 MW battery energy storage facility interconnecting at the Pana 138 kV substation in the Virginia area. The request was received on September 19, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 364 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$359,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Urban-Front Royal 345.0 kV
Contingency: International Paper Valliant Ok-Urban 345.0 kV
Loading: 100.7% (Rating: 1,485 MVA)
DFAX: 6.8%
Pre-project loading: 100.4%
2. Facility: American Substation-Urban 345.0 kV L
Contingency: Orchar Road-Fort Humbug 500.0 kV
Loading: 100.2% (Rating: 1,903 MVA)
DFAX: 8.3%
Pre-project loading: 99.9%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Urban-Front Royal 345.0 kV	Reconductor line	\$15,450,000	1% (\$171,000)
American-Urban 345.0 kV	Reconductor line	\$21,035,000	1% (\$188,000)
	Total allocated to project		\$359,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.